Assignment 1 : Time series

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Exercice 1 :

By definition:

$$\begin{aligned} Var(\bar{Y}_n) &= Var\left(\frac{1}{n}\sum_{t=1}^{n} Y_t\right) \\ &= \frac{1}{n^2} Var\left(\sum_{t=1}^{n} Y_t\right) \\ &= \frac{1}{n^2}\left(\sum_{t=1}^{n} Var(Y_t) + 2\sum_{1\le i<j\le n} cov(Y_i, Y_j)\right) \end{aligned}$$

by stationarity, $Var(Y_t) = \gamma_0$, $\forall t \in \mathbb{Z}$

futhermore, $cov(Y_i, Y_j) = \gamma_{|i-j|}$ by symetry of cov

let $k = |i-j|$, we know that γ_k is only a function of k so we can transform the double sums to a single one indexed by k